الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
2011-1432

College of Computing and Informatics

Computer Networks

IT351

Project-Phase 2

Deadline: Day 09/08/2026 @ 23:59

[Total Mark is 15]

Name: Layan Alghamdi ID: S240014014
Name: Alanoud Alfalah ID: S240061825
Name: ID:

Instructions:

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, convert text into image or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
  - Phase 1 (Mid project Progress Report)
  - Phase 2 (Final project Report)
- This report is for Phase 2. In this report, you should complete all the requirements for Phase 2.
- Each Report will be evaluated according to the marking criteria

Learning Outcome(s):

Demonstrate protocol configuration, network-addressing schemes and analyze packet transmission for the Local and Wide Area Networks.

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. **WAN 1 (Main Campus):**
    - Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
    - Each LAN has 3 devices (PCs or laptops) connected through a switch.
    - Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
    - Connect each LAN to a router (Router 1).
2. **WAN 2 (Branch Campus)**:
    - Contains LAN 3 (Library) and LAN 4 (Faculty Office).
    - Each LAN has 3 devices (PCs or laptops) connected through a switch.
    - Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
    - Connect each LAN to a router (Router 2).
3. **WAN Connection**:
    - Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
    - Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. **Assign IP Addresses**:
    - Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
        - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).

- LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).
- LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. **Router Configuration**:
    - Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
        - Router 1:
            - LAN 1 Interface: 172.16.10.1/24
            - LAN 2 Interface: 172.16.20.1/24
            - WAN Interface (Serial): 10.10.10.1/30
        - Router 2:
            - LAN 3 Interface: 172.16.30.1/24
            - LAN 4 Interface: 172.16.40.1/24
            - WAN Interface (Serial): 10.10.10.2/30
3. **Enable Routing**:
    - Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.

Phase 2 Requirements

Testing and Verification

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)

Written Reflection — End of Project Report (5 Marks)

The project report must include a Reflection section (300–500 words) as the last section before the Project Artifacts section. This section must be written in the students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method (e.g., static routing vs. RIP, your IP addressing scheme) over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

  - Live report document (with edit history): [link]
  - GitHub/GitLab repository (with commit history): [link]
  - Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The final report must be submitted as a PDF, together with the live document link, the repository link, and the Packet Tracer (.pkt) file.
- The edit history of the live document and the commit history of the repository must show incremental progress across both phases (Phase 1 and Phase 2).

Network Topology

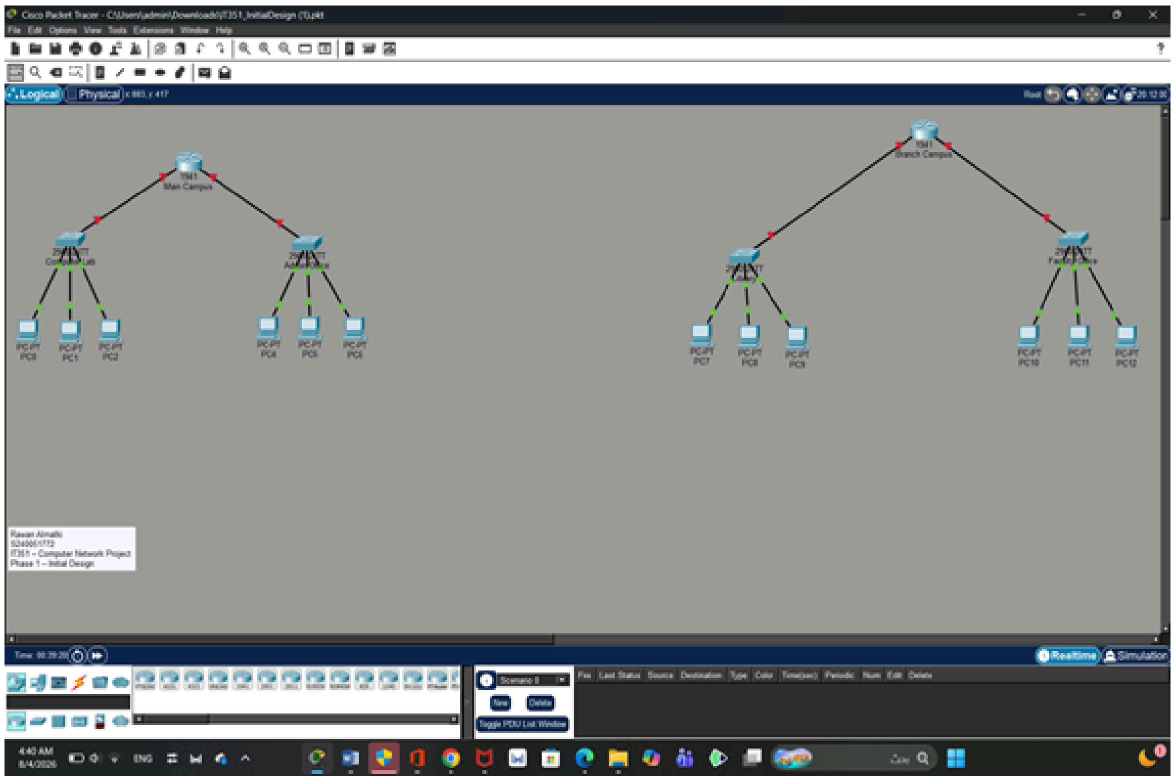


Figure 1. Initial Network Topology

Devices Used in the Network

Device Type	Model	Quantity
Router	Cisco 1941	2
Switch	Cisco 2960-24TT	4
PC	PC-PT	12

Commands Used to Change Device Names

Router (**Main Campus**):

enable

configure terminal

hostname MainCampus

end

copy running-config startup-config

Router (**Branch Campus**):

enable

configure terminal

hostname Branch Campus

end

copy running-config startup-config

Switch (**Computer Lab**) :

enable

configure terminal

hostname ComputerLab

end

copy running-config startup-config

Switch (**Admin Office**):

enable

configure terminal

hostname AdminOffice

end

copy running-config startup-config

Switch (**Library**):

enable

configure terminal

hostname Library

end

copy running-config startup-config

Switch (**Faculty Office**):

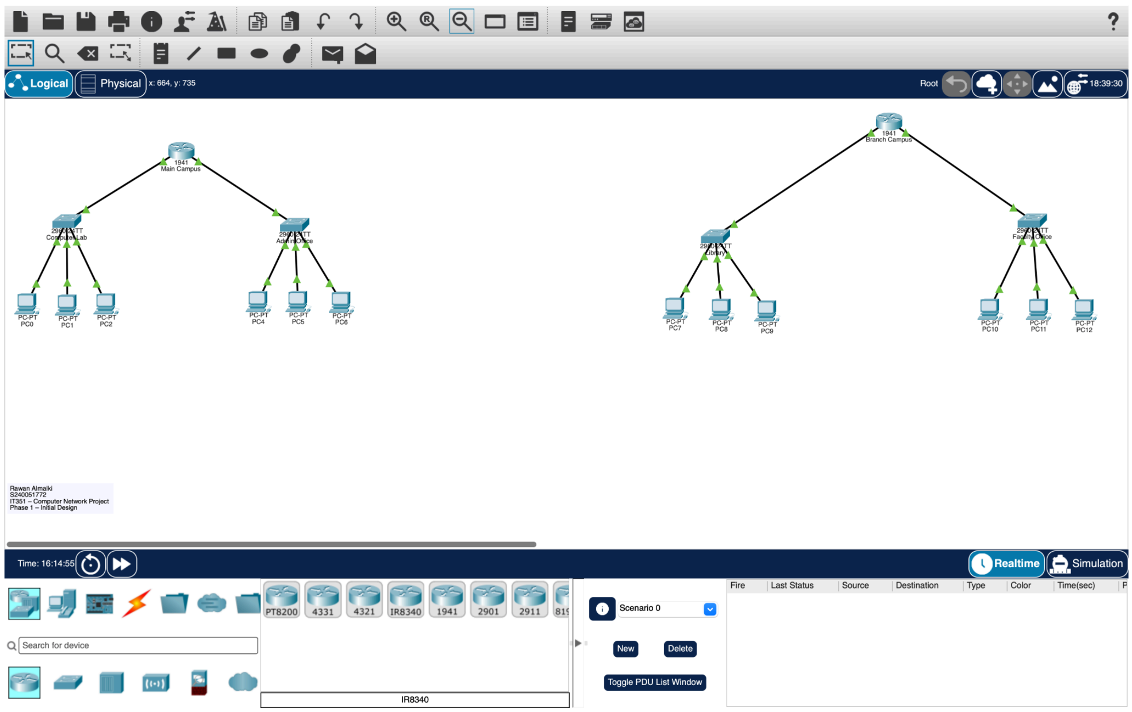
enable

configure terminal

hostname FacultyOffice

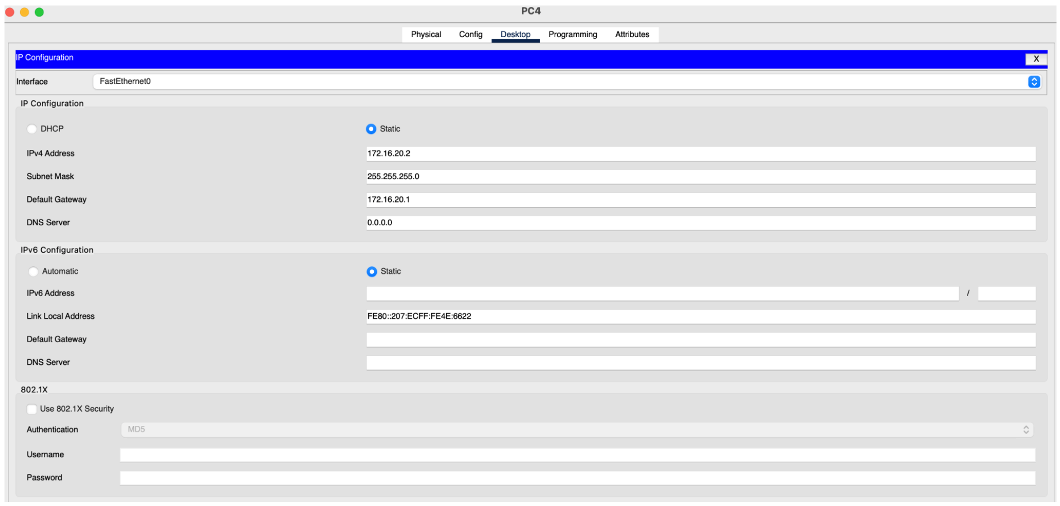
end

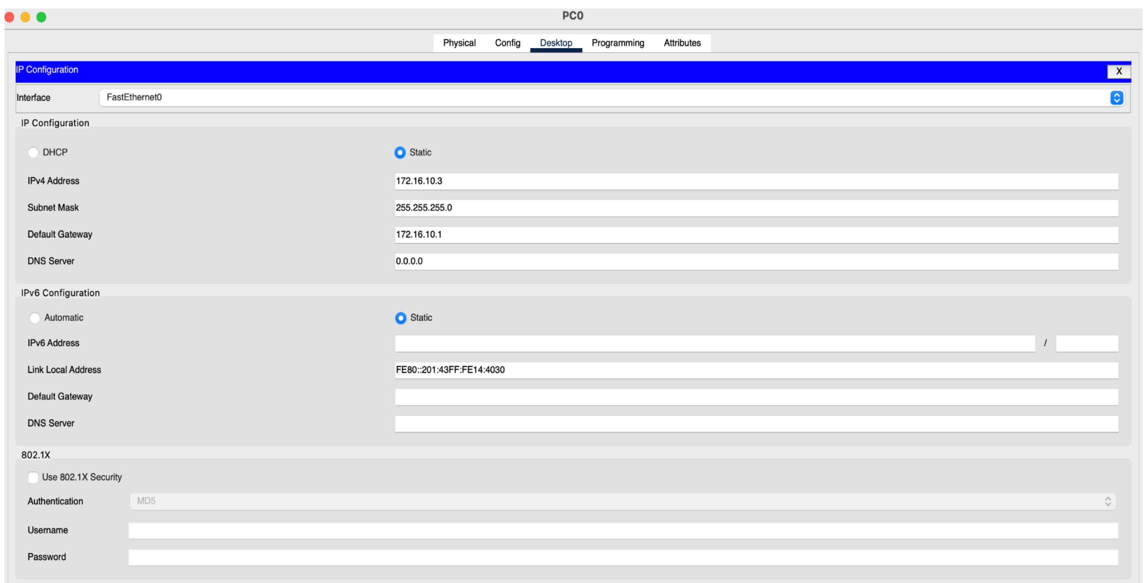
copy running-config startup-config



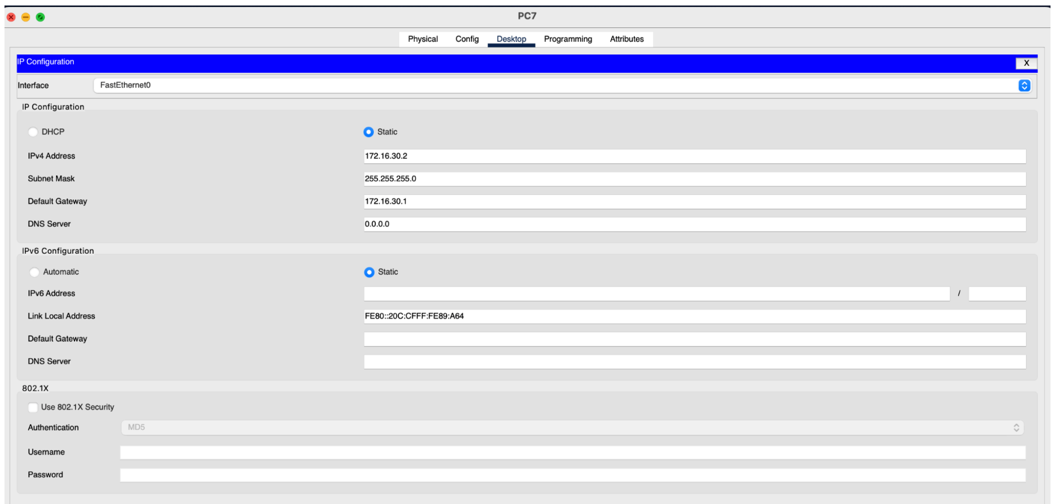
Connection	Cable Type	Reason
PC to Switch	Copper Straight-Through	Used to connect different types of devices (PC and Switch).
Switch to Router	Copper Straight-Through	Used to connect different types of devices (Switch and Router).

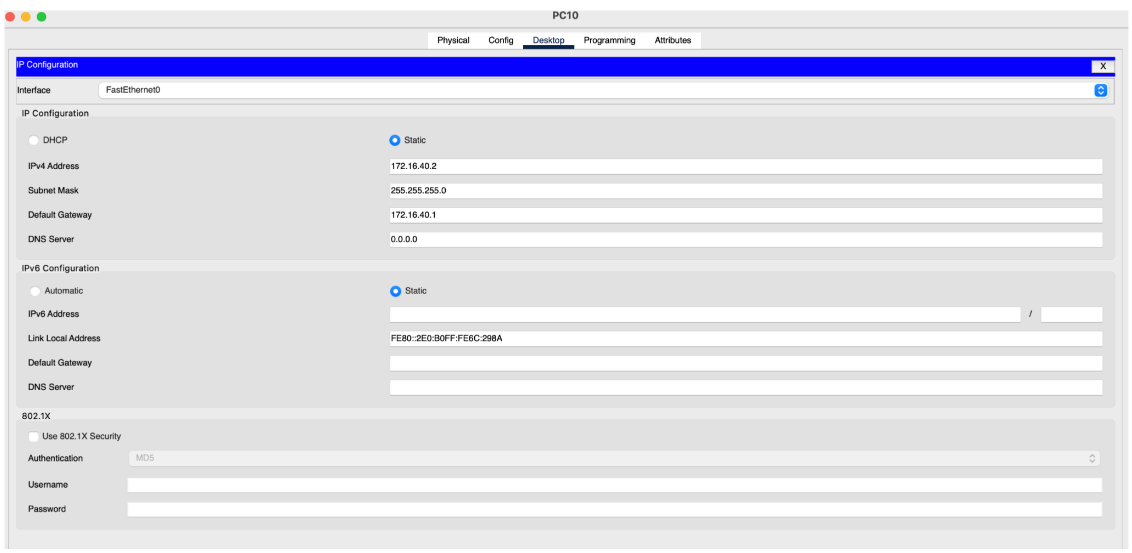
LAN	Network	Subnet Mask	Default Gateway
ComputerLab	172.16.10.0/24	255.255.255.0	172.16.10.1
AdminOffice	172.16.20.0/24	255.255.255.0	172.16.20.1
Library	172.16.30.0/24	255.255.255.0	172.16.30.1
FacultyOffice	172.16.40.0/24	255.255.255.0	172.16.40.1


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.20.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::207:ECFF:FE4E:6622
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

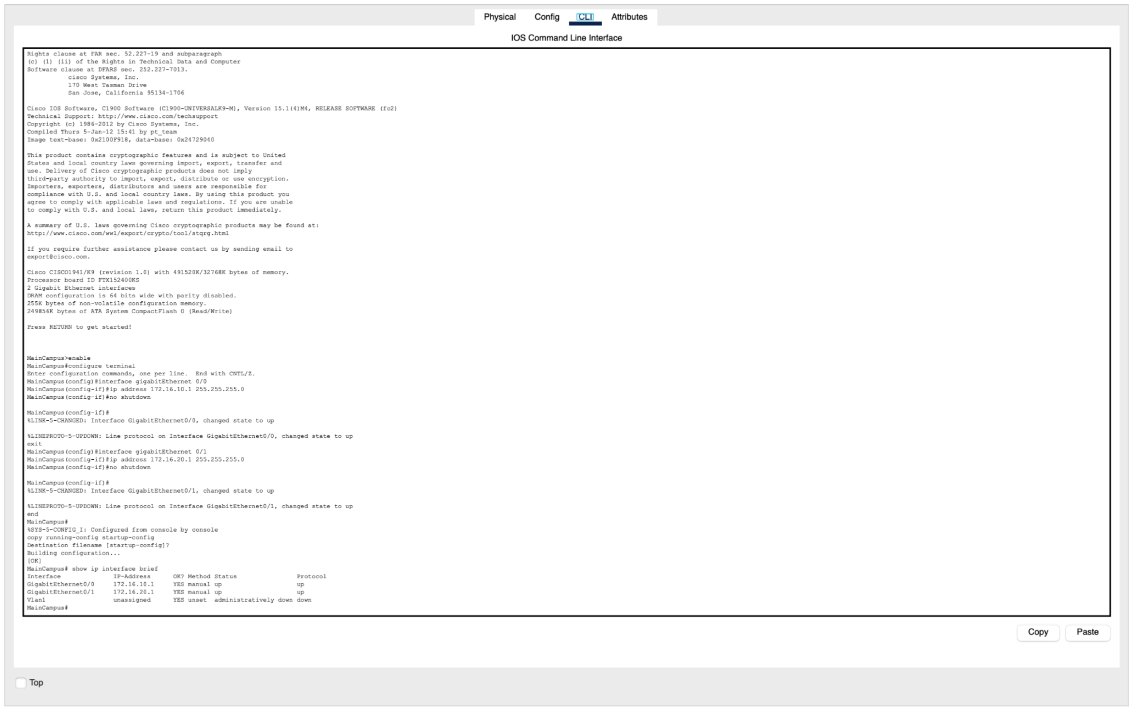

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.10.3
Subnet Mask
255.255.255.0
Default Gateway
172.16.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:43FF:FE14:4030
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق Heading
على النص الذي ترغب في أن يظهر هنا 1.





خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق Heading 1 على النص الذي ترغب في أن يظهر هنا.



enable

configure terminal

interface gigabitEthernet 0/0

ip address 172.16.10.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 172.16.20.1 255.255.255.0

no shutdown

end

copy running-config startup-config

enable

configure terminal

interface gigabitEthernet 0/0

ip address 172.16.30.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 172.16.40.1 255.255.255.0

no shutdown

end

copy running-config startup-config